

PRESENTED BY :

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SmartShift: Real-Time Shade
Tracking and Scheduling Model

PROJECT PROPOSAL

PROBLEM STATEMENT



WHY

Sun exposure affects ~2.4 billion outdoor workers globally ($\approx 70\%$ of the world's workforce), and this risk is rapidly increasing due to climate change and urbanization.

In the GCC and similar regions, temperatures often exceed 40°C for months, making traditional fixed schedules unsafe. People involved in outdoor work and daily commutes remain exposed to dangerous heat with limited guidance. Cities also lack accessible, data-driven tools to plan shade and manage heat at a micro-location level.

- **Problem:** Current systems rely on broad temperature or UV data, overlooking microclimates and temporary shade variations that could make certain areas safer at specific times. There is a clear need for a location and time-aware planning system that helps people and workers operate safely under varying heat conditions.
- **Goal:** To address these gaps with smart technology to minimize heat risks, improve comfort, and make it accessible for all outdoor users.

TARGET USERS

Outdoor & Field workers
construction crews, municipal workers & outdoor cleaners

Daily Commuters/Activity Participants
commuters, runners, cyclists, marathoners

Urban Planners & Developers
landscapers, urban designers, small-to-medium businesses, and households



KEY OBJECTIVES



Develop a low-cost tool that

Predicts sun movement and tracks shaded areas using 3D urban structures and sun movement calculations. Combines existing datasets that show the urban 3D geographic mapping of the sun movement and determines the shade accordingly.

- **Determines** optimal shift schedules, routes, and timings to minimize direct sun exposure
- **Provides** visual measures of heat and UV exposure levels
- **Recommends** ideal schedules and times for outdoor activities
- **Simulates** sun and shadow movement to optimize placement and design for sun exposure
- **Integrates** real-time and forecasted heat data to assess safe working conditions
- **Sends** alerts and recommendations during dangerous heat levels

KEY OUTCOMES



OUTDOOR & FIELD WORKERS

- **Health:** Reduce heat-related health risks and prevent heatstroke.
- **Well-being:** Increase worker satisfaction and well-being.
- **Productivity:** Boost productivity through safer, optimized scheduling.

DAILY COMMUTERS & ACTIVITY PARTICIPANTS

- **Safety:** Encourage safer participation and higher physical activity.
- **Comfort:** Enhance overall comfort & health during outdoor activities.
- **Efficiency:** Lower costs by optimizing event scheduling and space usage.

URBAN PLANNERS & DEVELOPERS

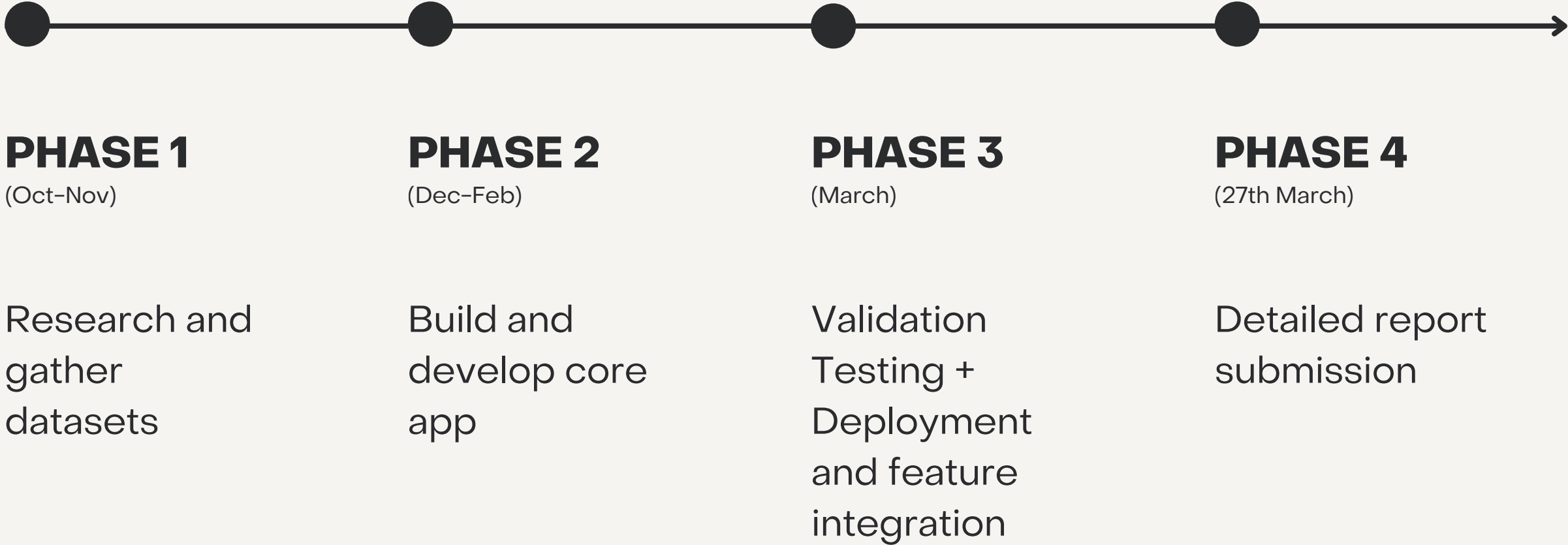
- **Sustainability:** Support sustainable planning with efficient shading and solar strategies.
- **Insight:** Enable data-driven decisions for city design and resource allocation.

KEY WORKSTREAMS



Workstream	Activity
Current State Assessment	• Assess existing tools & technologies that either already exist or can be used for the development of the model • Identify and engage key stakeholders to validate & refine the objectives
Target State Design	Developing the target model and architecture • Target 1: 3D urban mapping • Target 2: Solar and Shadow Mapping • Target 3: Heat Risk Predictor Model • Target 4: Scheduling Engine
Testing & Optimization	Field testing & model performance validation. Optimize based on feedback.
Solution Delivery	• Deliver a fully functional model. • Integrate all modules for end-to-end usability. • Ensure scalability and deployment readiness.
Documentation & Submission	Prepare final report. Submit project deliverables for review and evaluation.

PROJECT TIMELINE



PROJECT RESOURCES



DATA SOURCES

- Urban 3D Data: Mapbox API for 3D LOD2 building structures (based on OpenStreetMap for building footprints and heights).
- Solar Position Data: NOAA Solar Calculation algorithm to compute sun angles through the day in Python
- Heat and Weather Data: Open-Meteo SDK for Python for temperature, humidity, wind, and UV index. NASA Landsat or VIIRS for Land Surface Temperature (thermal imagery).
- Satellite Imagery: Google Earth Engine for integrating and visualizing thermal/spectral data.

FRAMEWORKS

- Computation & Modeling: Python, Pandas, OpenMeteo, NumPy, Scikit-learn, Joblib
- Optimization & Scheduling: OR-Tools, Pyomo (for route/schedule planning)
- Frameworks & Visualization: Flask API, Mapbox for interactive maps
- Database & APIs: Mapbox API for 3D Visualization, RestfulAPI for backend services, SQLite3 for schedule storage

RISK ASSESSMENT

<i>Risks</i>	<i>Likelihood</i>	<i>Impact</i>	Proposed Mitigation
Data accuracy — Incomplete or low-quality datasets (3D urban layers, and heat & weather data) may affect model reliability	High	High	Identify data gaps early; use validated public datasets; cross-check results with sample real-world data for accuracy.
Tight delivery timelines (MVP risk) — Limited time may constrain feature development and testing	High	High	Focus on core functionalities; limit MVP scope; hold weekly progress reviews to manage priorities.
Existing solution overlap — Similar tools may already exist, reducing differentiation	High	Medium	Conduct a market and technology review; emphasize innovation, integration, and localized application advantages.
Solution does not address real needs — Misalignment with end-user or stakeholder requirements	Low	Medium	Validate objectives with key stakeholders; support with data evidence (e.g., heat-related health risks); adjust model accordingly.
Integration complexity — Combining multiple data layers (3D urban, solar heatmaps, weather APIs) may introduce compatibility or performance issues	Medium	Medium	Develop modular architecture; test integration iteratively; use standardized data formats and APIs.

ISSUES LOG

Issues	<i>Issues & Impact Description</i>	<i>Severity</i>	Remedial Action(s)
Dataset is outdated (collected in 2020).	Outdated dataset may reduce the model’s relevance, accuracy, and ability to generalise to current structures and areas.	Medium	– Source a more recent dataset if available. – Augment the dataset with synthetic or simulated data – Perform exploratory analysis to identify outdated features and adjust preprocessing accordingly.
3D LOD2+ open source data difficult to obtain.	3D city models are too expensive and difficult to build from scratch due to lack of open source LOD2+ data available. Will impact the quality of the product significantly as this is a foundational dataset.	Very High	• Look into sample models, targeting community bounding boxes instead of whole city. • Build a community model in AutoCAD/Blender
3D city model developed in CAD incompatible with Python code manipulation.	Readily imports extrusions over 2D OSM map with average accuracy of building height estimates but unrealistic coordinate positions compared to real city, only estimated.	Hlgh	• Alternate machine learning approach to generate building details over extrusions and heights obtained from GeoFabrik. • Potential free-to-use APIs.

RELATED DOCUMENTS



LIST ALL APPLICABLE LINKS

- Include links to meeting recordings and password
- Add other links relevant to the project you're proposing